

Cloud Computing

A practical study reference with foundational concepts, examples, and common review questions.

1. Concepts and Service Models

Cloud computing provides on-demand network access to configurable resources with self-service provisioning, elasticity, pooling, and measured usage. It does not automatically mean cheaper, more secure, or more reliable.

IaaS provides infrastructure; PaaS manages more of the platform; SaaS delivers an application. Serverless abstracts server management, not the existence of servers. Public, private, hybrid, and multicloud deployments differ in control, skills, and operations.

2. Infrastructure and Responsibility

A region is a geographic deployment area; availability zones are isolated locations where offered. Compute includes virtual machines, containers, managed platforms, and functions. Object, block, and file storage have different access semantics.

Cloud security is shared: providers secure parts of the service while customers remain responsible for areas such as identity, data, configuration, and applications. Use least privilege, MFA, protected secrets, encryption, logging, and tested backups.

3. Reliability, Observability, and Cost

Design for failure using health checks, timeouts, bounded retries with jitter, redundancy where justified, and graceful degradation. Unbounded retries can amplify outages. Define service objectives and recovery goals.

Metrics, logs, and traces help explain behavior. Monitor user-facing latency, errors, and availability. Cost includes compute, storage, requests, data transfer, licensing, and support; tag resources, set budgets, and review idle capacity.

4. Common Questions

Scalability is the ability to handle growth; elasticity adjusts capacity as demand changes. A container packages an application process and dependencies but is not automatically a VM or complete security boundary. Replication copies data; sharding partitions it.